

Scalability Analysis of Hybrid WDM/TDM Passive Optical Networks: A Comparative Study of 4 Channel and 8 Channel Architectures Using OptiSystem Simulation

MD Asadullah
Computer Science & Engineering
University of Asia Pacific
Dhaka, Bangladesh
23101126@uap-bd.edu

Nabiha Afrin
Computer Science & Engineering
University of Asia Pacific
Dhaka, Bangladesh
23101125@uap-bd.edu

Tamim Iqbal
Computer Science & Engineering
University of Asia Pacific
Dhaka, Bangladesh
22201141@uap-bd.edu

Abstract—This paper presents a scalability study of hybrid Wavelength Division Multiplexing/Time Division Multiplexing Passive Optical Networks (WDM/TDM PON) through the design, simulation, and comparative performance analysis of two architectures: a 4-channel system and an 8-channel system, both implemented in OptiSystem 7.0. Each wavelength channel carries an independent 1 Gbps NRZ-OOK modulated stream, externally modulated through Mach-Zehnder Modulators with a 30 dB extinction ratio at 10 dBm launch power. The 4-channel system operates at carrier frequencies of 193.1, 193.2, 193.3, and 193.4 THz on the ITU-T DWDM grid, supporting eight end users at an aggregate capacity of 4 Gbps. The 8-channel system extends to 193.1–193.8 THz with 100 GHz spacing, supporting sixteen users at an aggregate capacity of 8 Gbps. Transmission performance is evaluated over standard single-mode fiber (SMF) with attenuation of 0.2 dB/km and chromatic dispersion of 16.75 ps/nm/km at fiber lengths of 10, 30, and 50 km. For the 4-channel system, Q-factors range from 385.82 at 10 km to 229.82 at 50 km with zero Bit Error Rate (BER) across all eight receiver branches. The 8-channel system maintains Q-factors between 245.37 and 135.70 across the same distances with BER of zero on every branch. The results demonstrate that channel-count scaling preserves error-free transmission while doubling both aggregate capacity and user density, and that the modest Q-factor reduction from 4-channel to 8-channel operation remains well above the industry $Q \geq 6$ threshold, establishing the hybrid WDM/TDM PON as a scalable candidate for next-generation optical access networks.

Index Terms—Passive Optical Network, Hybrid WDM/TDM PON, Scalability, Aggregate Capacity, Q-Factor, Bit Error Rate, Mach-Zehnder Modulator, OptiSystem, NRZ Modulation, Wavelength Utilization, Quality of Service

I. INTRODUCTION

The exponential growth of bandwidth-intensive applications including high-definition video streaming, cloud computing, Internet of Things (IoT), and emerging 5G mobile backhaul has placed unprecedented pressure on optical access networks worldwide. Bangladesh, as a rapidly developing nation with increasing broadband penetration, faces similar demands for

scalable, cost-effective, and high-capacity optical network infrastructure. Optical fiber, with its enormous low-loss spectral window spanning approximately 30 THz in the C-band and L-band, remains the only transmission medium capable of satisfying these requirements at acceptable cost per bit [1].

Passive Optical Networks (PON) have emerged as the dominant technology for broadband access due to their elimination of active components in the distribution network, reduced operational costs, and ability to serve multiple users over shared fiber infrastructure. Traditional Time Division Multiplexing PON (TDM-PON) architectures such as GPON and EPON have been widely deployed, but their fundamental shared-wavelength operation limits per-user throughput as subscriber counts increase [4]. Pure Wavelength Division Multiplexing PON (WDM-PON) overcomes this limitation by assigning dedicated wavelengths to user groups, yet the cost of wavelength-selective components at every Optical Network Unit (ONU) makes large-scale deployment economically prohibitive [6].

The hybrid WDM/TDM PON architecture addresses both limitations by combining the spectral efficiency of WDM with the cost-effectiveness of TDM. Multiple wavelength channels are transmitted simultaneously on the same fiber, with each channel shared among a group of users through time division multiplexing. This combination yields higher aggregate bandwidth, longer transmission reach, and improved quality of service compared to TDM-PON, while keeping component costs lower than pure WDM-PON [9]. Earlier work on small-scale hybrid systems with one or two wavelength channels established the feasibility of the architecture [10], but the scaling behavior of the hybrid PON as the number of wavelength channels grows has received comparatively limited attention in the simulation literature. As channel counts increase, several questions become important: how does the per-channel Q-factor degrade with additional wavelengths sharing

the same fiber, how is the optical power budget affected by larger multiplexers and demultiplexers, and how reliably does BER remain at acceptable levels as user density doubles?

This paper addresses these questions through a scalability-focused simulation study and makes the following contributions:

- 1) A complete OptiSystem 7.0 physical-layer simulation of a 4-channel hybrid WDM/TDM PON at 193.1–193.4 THz operating at 1 Gbps per channel, supporting eight end users at 4 Gbps aggregate capacity.
- 2) An extension to an 8-channel hybrid WDM/TDM PON at 193.1–193.8 THz with 100 GHz spacing, supporting sixteen end users at 8 Gbps aggregate capacity.
- 3) Comprehensive performance evaluation of both systems using Q-factor, minimum BER, eye height, and optical power measurements at 10, 30, and 50 km fiber lengths across every receiver branch.
- 4) A scalability comparison quantifying how aggregate capacity, user count, wavelength utilization, and Q-factor evolve as channel count doubles, providing design guidance for further scaling toward 16 and 32 wavelengths.

II. RELATED WORK

The field of passive optical networking has evolved rapidly over the past two decades, driven by increasing subscriber bandwidth demands and advances in optical component technology. This section reviews prior work on TDM-PON, WDM-PON, and hybrid WDM/TDM PON architectures that informs the scalability study presented here.

A. TDM-PON Architectures and Limitations

The foundational TDM-PON standards, GPON (ITU-T G.984) and EPON (IEEE 802.3ah), established the basic PON architecture of a tree topology with passive optical splitters, a single downstream wavelength at 1490 nm, and upstream transmission at 1310 nm using time-division multiple access. Koonen [4] provided an authoritative survey of FTTH technologies, comparing point-to-point, active, and passive optical networks on cost, capacity, reach, and operational complexity. The survey established that while TDM-PON offered the best cost-per-subscriber metric at moderate bandwidth levels, the shared-wavelength architecture would become a limiting factor as per-user demands grew. Uzar and Ünverdi [5] provided component-level analysis of OLT, ONU, and passive splitter characteristics that established parameter ranges used in subsequent simulation studies.

B. WDM-PON and Spectral Efficiency

Keiser [6] and Green [7] provided comprehensive treatments of FTTH concepts including WDM-PON system design, component selection, and performance characterization. These works established the theoretical capacity advantages of WDM-PON and identified the cost of wavelength-selective components as the primary obstacle to mass-market adoption. Rajalakshmi et al. [8] performed systematic simulation analysis comparing TDM-PON and WDM-PON performance

using different coding schemes for extended reach scenarios and demonstrated through OptiSystem simulations that WDM-based approaches consistently achieve higher Q-factors and lower BER compared to TDM-PON at fiber lengths exceeding 20 km. Their simulation methodology and parameter selection informed the approach adopted here.

C. Hybrid WDM/TDM PON

Jain and Therese [9] proposed and evaluated hybrid WDM/TDM PON systems using star topology in OptiSystem, demonstrating through simulation that the hybrid architecture achieves Q-factors exceeding 40 at 50 km and significantly outperforms TDM-PON counterparts in both Q-factor and BER metrics. Their work provided the foundational simulation framework that the present paper builds upon and extends with multi-channel scalability analysis. Koman and Ünverdi [10] performed detailed comparative analysis of TDM-PON and small-scale hybrid WDM/TDM PON architectures at 193.1 THz using OptiSystem 16.0. They reported Q-factors of 450.78 at 10 km and 45.48 at 50 km for a two-channel hybrid system, and confirmed that TDM-PON produces non-zero BER at 50 km while the hybrid system maintains BER of zero. Their results establish the baseline against which the multi-channel scaling reported in this paper is interpreted.

D. Dynamic Bandwidth Allocation

Dhaini et al. [11] investigated dynamic wavelength and bandwidth allocation (DWBA) algorithms for hybrid TDM/WDM EPON networks, demonstrating that intelligent dynamic allocation substantially reduces mean packet delay and improves quality of service differentiation compared to static allocation. Their work established the QoS framework that motivates the channel scaling study reported here, since DWBA gains grow with the number of available wavelengths.

Despite this body of literature, few prior publications have reported complete physical-layer OptiSystem simulations of hybrid WDM/TDM PON architectures with side-by-side scaling from four wavelength channels to eight, with detailed per-branch performance evaluation across multiple fiber lengths. The present paper addresses that gap directly.

III. PROPOSED SCALABLE HYBRID WDM/TDM PON ARCHITECTURE

A. Architectural Overview

The proposed scalable hybrid WDM/TDM PON architecture is structured in a modular form that supports both the 4-channel and 8-channel configurations studied in this paper. Each configuration shares a common building-block pattern: N_λ wavelength-distinct transmitter chains feed into an $N_\lambda \times 1$ WDM multiplexer, which launches the combined optical signal into standard single-mode fiber. At the receiver end, a $1 \times N_\lambda$ WDM demultiplexer separates the wavelengths onto independent ports, and a 1×2 power splitter on each port distributes the signal to two end users. The architecture therefore supports $2N_\lambda$ end users in total for N_λ wavelength

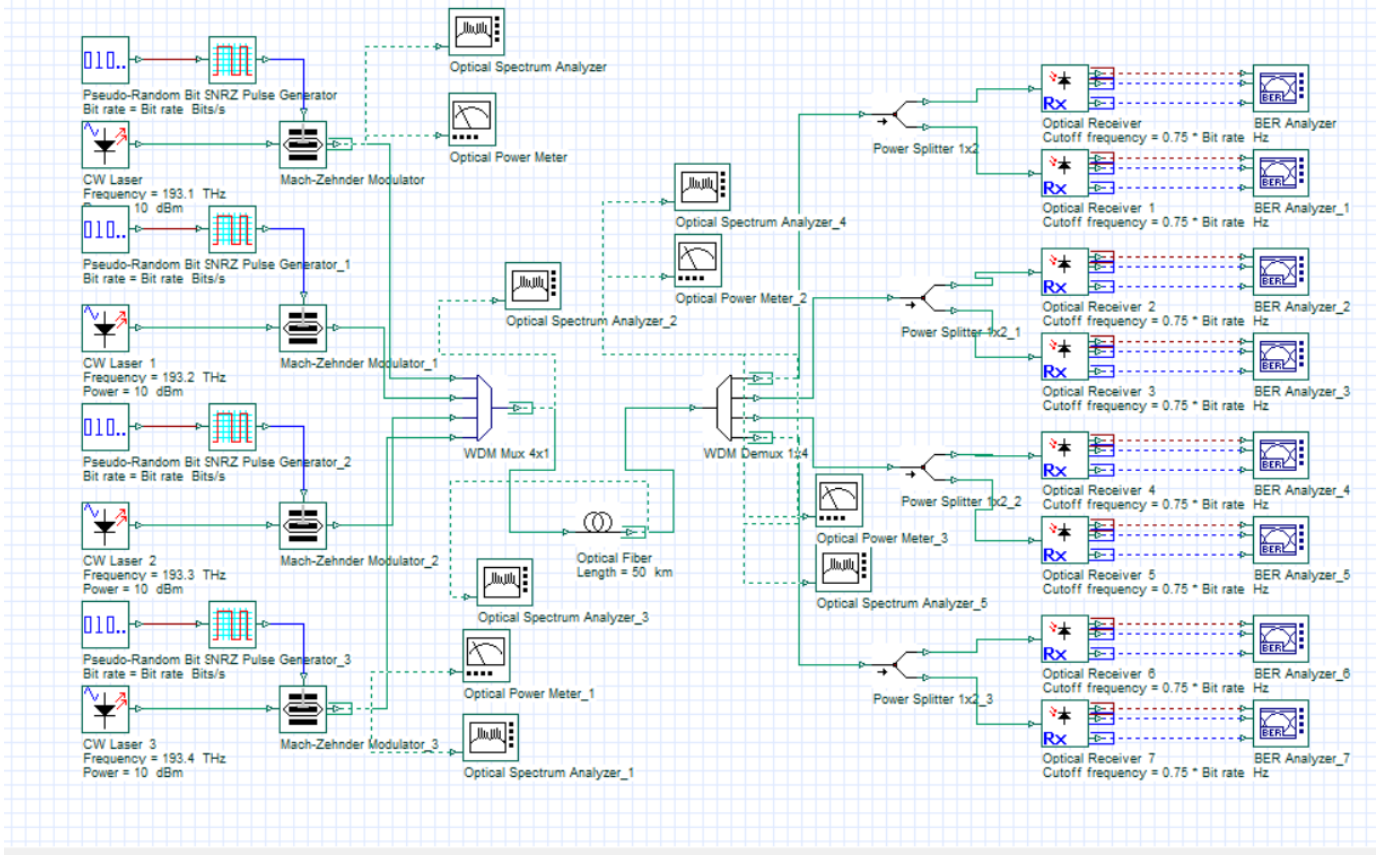


Fig. 1. Proposed 4-Channel Hybrid WDM/TDM PON System Topology in OptiSystem 7.0. Four transmitters at 193.1, 193.2, 193.3, 193.4 THz feed a 4×1 WDM MUX, traverse SMF, and are demultiplexed to eight receivers through 1×2 power splitters.

channels, with aggregate capacity $C_{\text{total}} = N_{\lambda} \cdot C_{\text{channel}}$ where $C_{\text{channel}} = 1$ Gbps in this study.

B. 4-Channel Architecture

The 4-channel hybrid WDM/TDM PON, shown in Fig. 1, employs four wavelength channels at 193.1, 193.2, 193.3, and 193.4 THz on the ITU-T DWDM grid with 100 GHz spacing. Four independent 1 Gbps NRZ-OOK transmitter chains feed a 4×1 WDM multiplexer. After fiber transmission, a 1×4 WDM demultiplexer recovers the four channels, each of which is split through a 1×2 power splitter to serve two end users. Eight receiver branches are therefore monitored simultaneously, yielding an aggregate capacity of 4 Gbps.

C. 8-Channel Architecture

The 8-channel hybrid WDM/TDM PON, shown in Fig. 2, doubles the wavelength count to eight channels spanning 193.1–193.8 THz with 100 GHz spacing. An integrated WDM transmitter with eight output ports replaces the four discrete CW-laser and Mach-Zehnder transmitter chains used in the 4-channel design; this choice preserves the per-channel modulation characteristics (NRZ-OOK at 10 dBm, 30 dB extinction ratio) while keeping the topology compact. An 8×1 WDM multiplexer combines the eight channels onto a single fiber. At the receiver, a 1×8 WDM demultiplexer separates the

wavelengths, and a 1×2 power splitter on each output yields sixteen receiver branches, supporting sixteen end users at an aggregate capacity of 8 Gbps.

D. Transmitter Section

Each transmitter chain in the 4-channel system comprises a pseudo-random bit sequence (PRBS) generator producing a 1 Gbps binary stream with seventh-order polynomial sequence of 128 bits, an NRZ pulse generator with exponential pulse shaping and rise/fall time of 0.05, a CW laser at 10 dBm and 10 MHz linewidth, and a Mach-Zehnder Modulator (MZM) with 30 dB extinction ratio and symmetry factor of -1 . The four lasers are tuned to 193.1, 193.2, 193.3, and 193.4 THz, respectively. The 8-channel system uses the OptiSystem WDM Transmitter component configured for eight output ports starting at 193.1 THz with 100 GHz frequency spacing, which integrates the equivalent functionality of eight laser/PRBS/NRZ/MZM chains while keeping per-channel parameters identical.

E. Multiplexing Stage

For the 4-channel system, the WDM 4×1 multiplexer is configured with passband centers at 193.1, 193.2, 193.3, and 193.4 THz, a second-order Bessel filter, 10 GHz bandwidth per channel, and 100 dB extinction depth. The 8-channel system

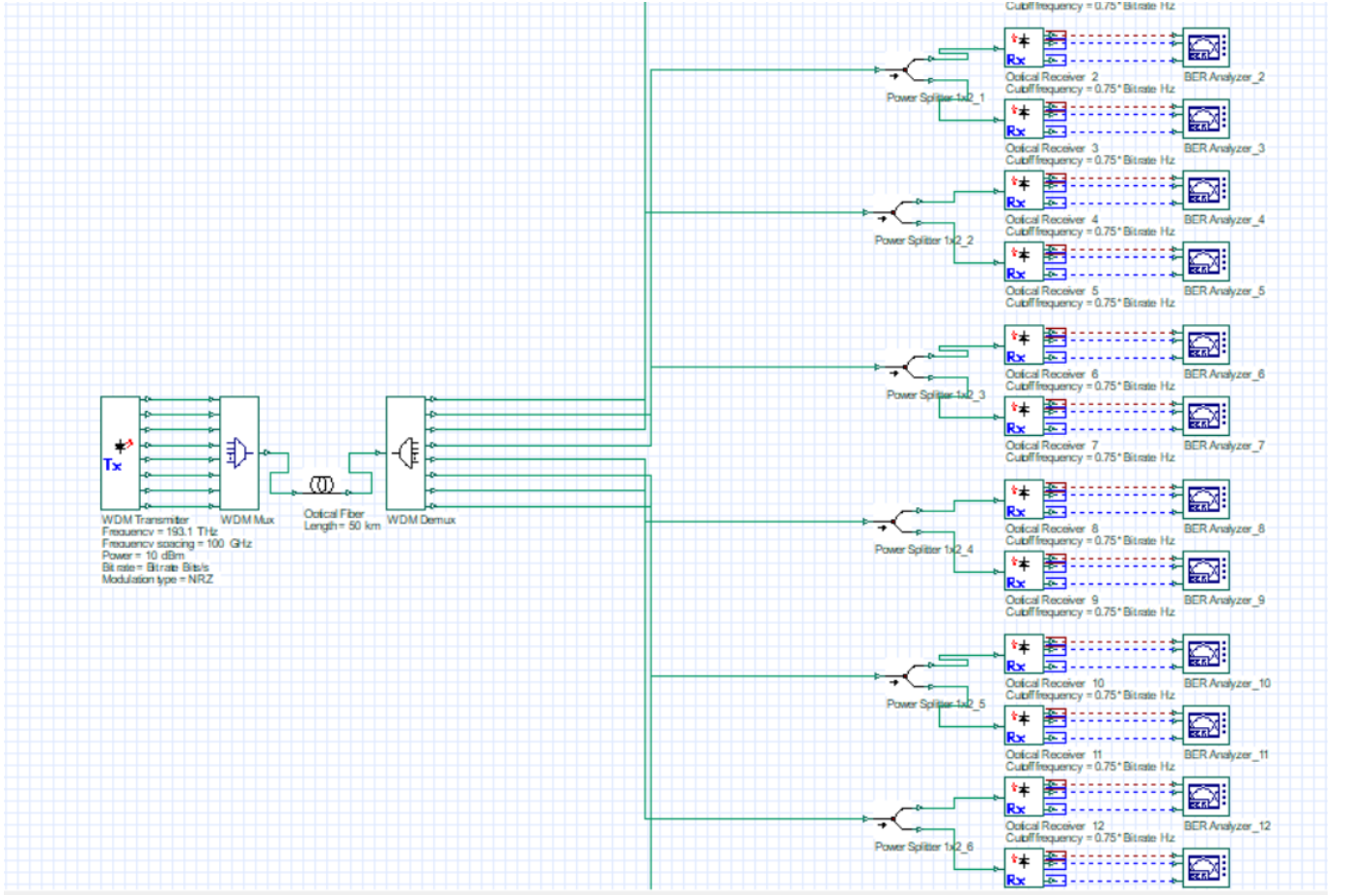


Fig. 2. Proposed 8-Channel Hybrid WDM/TDM PON System Topology in OptiSystem 7.0. An integrated 8-port WDM transmitter spanning 193.1–193.8 THz feeds an 8×1 WDM MUX; sixteen receivers are served through a 1×8 DEMUX and eight 1×2 power splitters.

uses an equivalent 8×1 multiplexer with the same filter design extended to passband centers at 193.1–193.8 THz. Optical Spectrum Analyzers (OSAs) are placed after each transmitter output and after each multiplexer output to verify spectral integrity.

F. Transmission Span

Both systems use standard ITU-T G.652 single-mode fiber with attenuation $\alpha = 0.2$ dB/km, chromatic dispersion $D = 16.75$ ps/nm/km, dispersion slope 0.075 ps/nm²/km, effective area $80 \mu\text{m}^2$, and nonlinear refractive index $n_2 = 2.6 \times 10^{-20}$ m²/W. Fiber lengths of 10, 30, and 50 km are evaluated independently for both architectures.

G. Receiver Section

A WDM $1 \times N_\lambda$ demultiplexer separates the wavelength channels onto independent output ports, each of which connects to a 1×2 power splitter feeding two PIN photodetector-based optical receivers with responsivity 1 A/W, cutoff frequency 750 MHz, and reference bit rate 1 Gbps. The 4-channel system therefore presents eight receiver branches and the 8-channel system presents sixteen. Each branch is monitored by a BER Analyzer configured for Gaussian noise statistics.

IV. SIMULATION SETUP AND PARAMETERS

The simulation is implemented in OptiSystem 7.0 with the global layout configured to set bit rate mode. Table I summarizes the parameter set, which is identical for both the 4-channel and 8-channel configurations except for the channel count.

V. MATHEMATICAL PERFORMANCE MODEL

A. Q-Factor

The Q-factor quantifies signal quality at the receiver decision instant and is defined as

$$Q = \frac{\mu_1 - \mu_0}{\sigma_1 + \sigma_0} \quad (1)$$

where μ_1 and μ_0 are the mean signal amplitudes for logic ‘1’ and logic ‘0’, and σ_1 , σ_0 are the corresponding noise standard deviations [10].

B. Bit Error Rate

The minimum BER follows from the Q-factor under the Gaussian noise approximation:

$$\text{BER} = \frac{1}{2} \text{erfc}\left(\frac{Q}{\sqrt{2}}\right) \quad (2)$$

TABLE I
COMPLETE SIMULATION PARAMETER SET

Parameter	Value
Bit Rate (per channel)	1.00×10^9 bps
Sequence Length	128 bits
Samples per Bit	64
Number of Samples	8192
Sample Rate	6.40×10^{10} Hz
Time Window	1.28×10^{-7} s
4-Channel Frequencies	193.1, 193.2, 193.3, 193.4 THz
8-Channel Frequencies	193.1–193.8 THz (100 GHz spacing)
Launch Power	10 dBm
Laser Linewidth	10 MHz
MZM Extinction Ratio	30 dB
MZM Symmetry Factor	–1
NRZ Rise/Fall Time	0.05
PRBS Order	7
Fiber Attenuation	0.2 dB/km
Fiber Dispersion	16.75 ps/nm/km
Dispersion Slope	0.075 ps/nm ² /km
Effective Area	80 μm^2
n_2	2.6×10^{-20} m ² /W
Fiber Lengths	10, 30, 50 km
WDM MUX/DEMUX Filter	Bessel, Order 2
WDM MUX/DEMUX BW	10 GHz per channel
Receiver Type	PIN Photodetector
Receiver Responsivity	1 A/W
Receiver Cutoff Freq.	750 MHz
Power Splitter Loss	0 dB
BER Algorithm	Gaussian

The industry standard requires $Q \geq 6$ corresponding to BER $\leq 10^{-9}$ for an acceptable optical link.

C. Eye Height

The eye height quantifies the vertical opening of the eye diagram:

$$EH = (\mu_1 - 3\sigma_1) - (\mu_0 + 3\sigma_0) \quad (3)$$

D. Power Budget Including MUX/DEMUX Loss

The received optical power after fiber transmission is

$$P_{\text{rx}} [\text{dBm}] = P_{\text{tx}} - \alpha L - L_{\text{mux}}(N_\lambda) - L_{\text{demux}}(N_\lambda) - L_{\text{split}} \quad (4)$$

where $L_{\text{mux}}(N_\lambda)$ and $L_{\text{demux}}(N_\lambda)$ depend on the channel count because larger multiplexers introduce slightly higher insertion loss for the same passband design.

E. Chromatic Dispersion Pulse Broadening

Temporal broadening of an NRZ pulse due to chromatic dispersion is

$$\Delta\tau = D \cdot \Delta\lambda \cdot L \quad (5)$$

For $B = 1$ Gbps, bit period $T_b = 1$ ns, and modulated linewidth $\Delta\lambda \approx 0.05$ nm, at 50 km

$$\Delta\tau = 16.75 \times 0.05 \times 50 \approx 41.9 \text{ ps}, \quad (6)$$

which represents approximately 4.2% of the bit period and confirms that dispersion-induced inter-symbol interference remains negligible at the simulated bit rate.

F. Aggregate Capacity and User Scaling

The total aggregate capacity scales linearly with channel count:

$$C_{\text{total}} = N_\lambda \times C_{\text{channel}} \quad (7)$$

yielding 4 Gbps for the 4-channel system and 8 Gbps for the 8-channel system. Supported user count is

$$N_{\text{users}} = 2N_\lambda \quad (8)$$

under the present 1×2 power-splitter configuration, giving eight users for the 4-channel system and sixteen for the 8-channel system.

VI. RESULTS AND DISCUSSION

A. Optical Power Analysis

Table II summarizes representative optical power measurements at the WDM MUX output and at the DEMUX output for both systems at all three fiber lengths. The MUX output power remains near +6.8 dBm for both architectures, confirming that combining additional wavelengths does not significantly reduce per-channel launch power within the simulation. After fiber transmission and demultiplexing, the per-port power follows the expected linear attenuation of 0.2 dB/km, with the 8-channel system showing slightly lower DEMUX-port power than the 4-channel system at each distance due to the additional insertion loss of the 8×1 multiplexing stage.

TABLE II
OPTICAL POWER AT KEY STAGES FOR BOTH ARCHITECTURES (dBm)

Measurement Point	10 km	30 km	50 km
4-ch MUX output (Ch1)	6.78	6.78	6.78
4-ch MUX output (Ch2)	6.85	6.85	6.85
4-ch DEMUX Port 1	4.72	0.72	-3.28
4-ch DEMUX Port 2	4.79	0.79	-3.21
8-ch DEMUX Port 1	3.55	-0.50	-4.81
8-ch DEMUX Port 2	3.49	-0.55	-4.85

Fig. 3 shows the OSA output at the 4-channel MUX, where the four wavelength peaks at 193.1–193.4 THz are clearly resolved with the expected 100 GHz spacing. Fig. 4 shows the corresponding OSA output at the 8-channel MUX, where all eight wavelengths from 193.1 to 193.8 THz are distinctly visible with comparable peak powers and clean inter-channel separation, confirming that the 8-channel design preserves spectral integrity.

B. 4-Channel Performance at 10 km

Table III reports the Q-factor, minimum BER, and eye height for all eight receiver branches of the 4-channel system at 10 km. Every branch achieves zero BER, and the Q-factor ranges from 247.63 to 385.82, well above the $Q \geq 6$ acceptance threshold. The receivers attached to the outer channels (193.1 and 193.4 THz) exhibit the highest Q-factors, while the inner channels (193.2 and 193.3 THz) show slightly lower values due to marginal spectral overlap with neighboring passbands.

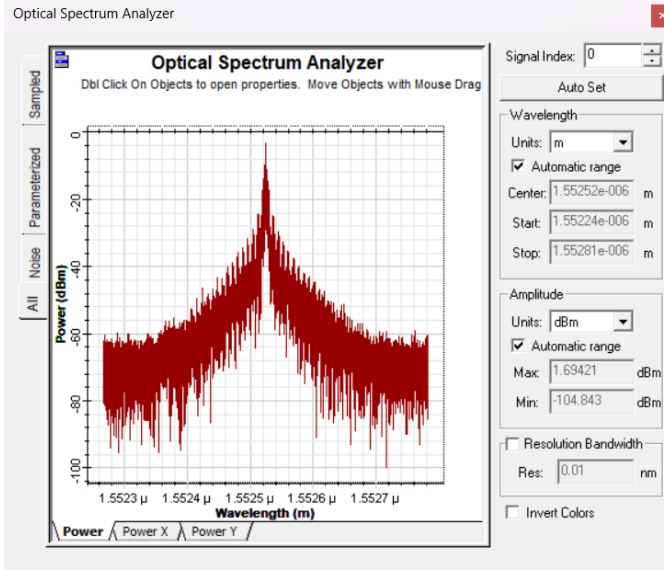


Fig. 3. Optical Spectrum Analyzer output after the 4-channel WDM MUX showing four wavelength peaks at 193.1, 193.2, 193.3, and 193.4 THz.

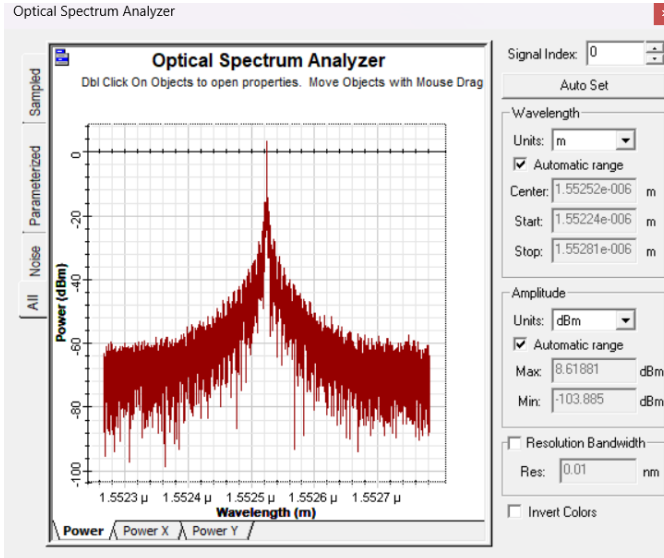


Fig. 4. Optical Spectrum Analyzer output after the 8-channel WDM MUX showing eight wavelength peaks at 193.1–193.8 THz with 100 GHz spacing.

TABLE III
4-CHANNEL BER ANALYZER RESULTS AT 10 KM

Receiver	Q-Factor	Min BER	Eye Height
Rx 1 (193.1 THz)	385.82	0	0.003118
Rx 2 (193.1 THz)	378.21	0	0.003118
Rx 3 (193.2 THz)	247.63	0	0.003109
Rx 4 (193.2 THz)	248.22	0	0.003109
Rx 5 (193.3 THz)	247.88	0	0.003100
Rx 6 (193.3 THz)	249.79	0	0.003100
Rx 7 (193.4 THz)	360.71	0	0.003120
Rx 8 (193.4 THz)	357.53	0	0.003120

C. 4-Channel Performance at 30 km

Table IV reports the corresponding results at 30 km. The Q-factor reduces moderately, with values from 267.98 to 347.61,

and BER remains zero on every branch. Eye height drops to approximately 0.00124, consistent with the reduced optical power at the photodetector after 30 km of fiber attenuation.

TABLE IV
4-CHANNEL BER ANALYZER RESULTS AT 30 KM

Receiver	Q-Factor	Min BER	Eye Height
Rx 1 (193.1 THz)	347.61	0	0.001241
Rx 2 (193.1 THz)	330.58	0	0.001241
Rx 3 (193.2 THz)	305.25	0	0.001240
Rx 4 (193.2 THz)	294.01	0	0.001240
Rx 5 (193.3 THz)	267.98	0	0.001236
Rx 6 (193.3 THz)	275.98	0	0.001236
Rx 7 (193.4 THz)	341.43	0	0.001241
Rx 8 (193.4 THz)	340.95	0	0.001241

D. 4-Channel Performance at 50 km

Table V reports the results at 50 km. Q-factors remain in the range 229.82–278.18, BER stays at zero on all eight branches, and eye height drops to approximately 4.93×10^{-4} . The Q-factor reduction from 10 km to 50 km is approximately a factor of 1.7, consistent with the linear fiber attenuation model and confirming that dispersion-induced impairments remain negligible at 1 Gbps over 50 km of standard SMF.

TABLE V
4-CHANNEL BER ANALYZER RESULTS AT 50 KM

Receiver	Q-Factor	Min BER	Eye Height
Rx 1 (193.1 THz)	278.18	0	0.000493
Rx 2 (193.1 THz)	247.26	0	0.000492
Rx 3 (193.2 THz)	237.81	0	0.000492
Rx 4 (193.2 THz)	238.10	0	0.000492
Rx 5 (193.3 THz)	229.82	0	0.000491
Rx 6 (193.3 THz)	234.23	0	0.000491
Rx 7 (193.4 THz)	269.20	0	0.000493
Rx 8 (193.4 THz)	276.05	0	0.000493

A representative eye diagram for the 4-channel system at 50 km is shown in Fig. 5. The eye remains visibly open with clear logic rails and stable decision thresholds, consistent with the reported Q-factor and zero BER.

E. 8-Channel Performance at 10 km

Table VI reports the Q-factor, minimum BER, and eye height for all sixteen receiver branches of the 8-channel system at 10 km. Doubling the channel count from four to eight reduces the achieved Q-factor relative to the 4-channel system at the same distance, with values ranging from 135.70 to 245.37. BER remains zero on every branch, and the Q-factor margin above the $Q \geq 6$ threshold is still very large.

F. 8-Channel Performance at 30 km

Table VII reports the 8-channel results at 30 km. Q-factors remain healthy, ranging from 161.45 to 253.18 with BER of zero on every branch. The mid-band channels (193.3–193.5 THz) again record the lowest Q-factors because they accumulate inter-channel spectral interaction from both lower and upper neighbors.

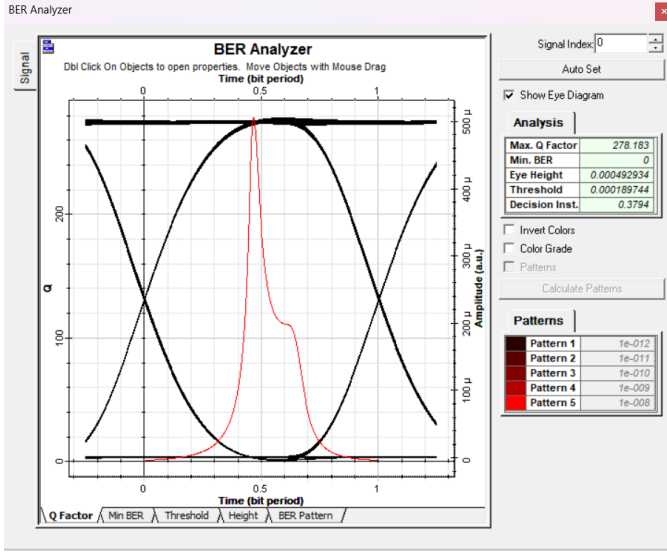


Fig. 5. Representative Eye Diagram for the 4-channel hybrid WDM/TDM PON at 50 km, showing a clearly open eye with stable logic rails.

TABLE VI
8-CHANNEL BER ANALYZER RESULTS AT 10 KM

Receiver	Q-Factor	Min BER	Eye Height
Rx 1 (193.1 THz)	245.37	0	0.002801
Rx 2 (193.1 THz)	243.91	0	0.002801
Rx 3 (193.2 THz)	189.91	0	0.002792
Rx 4 (193.2 THz)	189.98	0	0.002793
Rx 5 (193.3 THz)	158.33	0	0.002784
Rx 6 (193.3 THz)	157.08	0	0.002784
Rx 7 (193.4 THz)	138.35	0	0.002770
Rx 8 (193.4 THz)	137.32	0	0.002770
Rx 9 (193.5 THz)	135.78	0	0.002776
Rx 10 (193.5 THz)	136.59	0	0.002776
Rx 11 (193.6 THz)	173.64	0	0.002785
Rx 12 (193.6 THz)	171.78	0	0.002785
Rx 13 (193.7 THz)	175.08	0	0.002791
Rx 14 (193.7 THz)	173.95	0	0.002790
Rx 15 (193.8 THz)	208.62	0	0.002792
Rx 16 (193.8 THz)	204.75	0	0.002791

TABLE VII
8-CHANNEL BER ANALYZER RESULTS AT 30 KM

Receiver	Q-Factor	Min BER	Eye Height
Rx 1 (193.1 THz)	253.18	0	0.001116
Rx 2 (193.1 THz)	250.53	0	0.001116
Rx 3 (193.2 THz)	221.97	0	0.001114
Rx 4 (193.2 THz)	235.78	0	0.001115
Rx 5 (193.3 THz)	175.75	0	0.001111
Rx 6 (193.3 THz)	173.33	0	0.001111
Rx 7 (193.4 THz)	167.67	0	0.001108
Rx 8 (193.4 THz)	167.03	0	0.001107
Rx 9 (193.5 THz)	161.45	0	0.001106
Rx 10 (193.5 THz)	165.99	0	0.001107
Rx 11 (193.6 THz)	225.17	0	0.001115
Rx 12 (193.6 THz)	241.12	0	0.001116
Rx 13 (193.7 THz)	212.71	0	0.001114
Rx 14 (193.7 THz)	212.84	0	0.001114
Rx 15 (193.8 THz)	221.81	0	0.001112
Rx 16 (193.8 THz)	221.78	0	0.001112

G. 8-Channel Performance at 50 km

Table VIII reports the 8-channel results at 50 km. Across all sixteen branches, BER remains zero and Q-factors stay between 135.78 and 208.62. The eye height contracts to roughly 4.4×10^{-4} , slightly below the 4-channel value at the same distance due to the additional insertion loss of the larger multiplexer. The link still operates with more than $20\times$ the Q-factor margin required by the industry threshold.

TABLE VIII
8-CHANNEL BER ANALYZER RESULTS AT 50 KM

Receiver	Q-Factor	Min BER	Eye Height
Rx 1 (193.1 THz)	188.50	0	0.000442
Rx 2 (193.1 THz)	192.96	0	0.000443
Rx 3 (193.2 THz)	193.70	0	0.000443
Rx 4 (193.2 THz)	185.96	0	0.000442
Rx 5 (193.3 THz)	159.65	0	0.000441
Rx 6 (193.3 THz)	154.92	0	0.000441
Rx 7 (193.4 THz)	147.76	0	0.000440
Rx 8 (193.4 THz)	147.30	0	0.000440
Rx 9 (193.5 THz)	135.70	0	0.000439
Rx 10 (193.5 THz)	139.36	0	0.000439
Rx 11 (193.6 THz)	177.70	0	0.000442
Rx 12 (193.6 THz)	188.01	0	0.000442
Rx 13 (193.7 THz)	176.00	0	0.000442
Rx 14 (193.7 THz)	175.20	0	0.000442
Rx 15 (193.8 THz)	170.68	0	0.000442
Rx 16 (193.8 THz)	192.30	0	0.000443

A representative eye diagram for the 8-channel system at 50 km is shown in Fig. 6. Compared with the 4-channel eye at the same distance, the 8-channel eye is slightly narrower in amplitude but remains clearly open, supporting the zero BER result on every receiver branch.

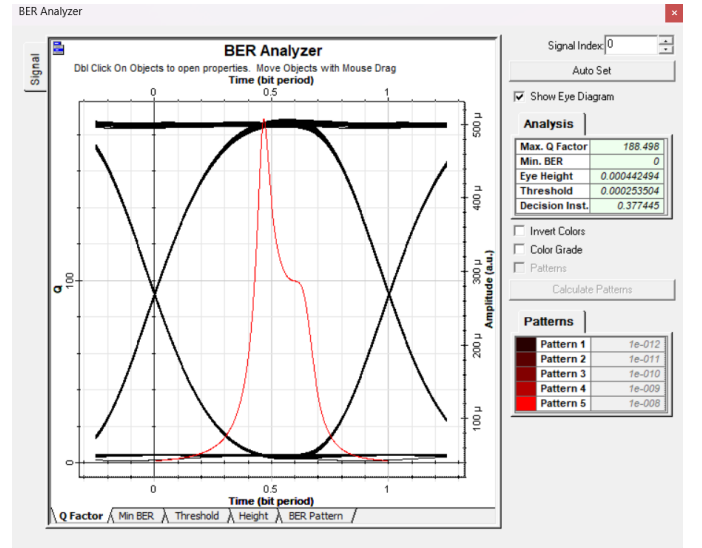


Fig. 6. Representative Eye Diagram for the 8-channel hybrid WDM/TDM PON at 50 km, showing a narrower but still clearly open eye.

H. Q-Factor Trend with Fiber Length

Fig. 7 plots the per-receiver Q-factor as a function of fiber length for both architectures. Two trends are apparent. First,

within each architecture the Q-factor decreases gradually with distance, consistent with the linear fiber attenuation model and confirming that dispersion remains a secondary impairment at 1 Gbps. Second, the 8-channel system operates at consistently lower per-branch Q-factor than the 4-channel system, but the gap narrows at longer distances because both systems become attenuation-limited rather than channel-spacing-limited.

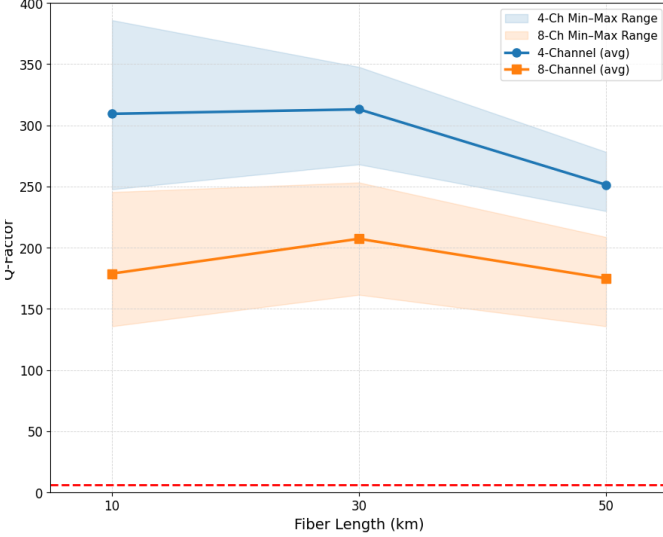


Fig. 7. Q-Factor vs Fiber Length for the 4-channel (8 receiver branches) and 8-channel (16 receiver branches) hybrid WDM/TDM PON architectures.

VII. COMPARATIVE SCALABILITY ANALYSIS

Table IX summarizes the side-by-side comparison between the 4-channel and 8-channel architectures. The headline result is that doubling the wavelength count from four to eight exactly doubles both aggregate capacity (from 4 to 8 Gbps) and supported user count (from 8 to 16), while keeping BER at zero on every receiver branch across the entire 10–50 km range.

TABLE IX
COMPARATIVE SCALABILITY: 4-CHANNEL VS 8-CHANNEL HYBRID WDM/TDM PON

Parameter	4-Channel	8-Channel
Wavelengths	4	8
Frequency Range	193.1–193.4 THz	193.1–193.8 THz
Aggregate Capacity	4 Gbps	8 Gbps
Users Supported	8	16
Min Q-Factor @ 10 km	247.63	135.70
Max Q-Factor @ 10 km	385.82	245.37
Min Q-Factor @ 30 km	267.98	161.45
Max Q-Factor @ 30 km	347.61	253.18
Min Q-Factor @ 50 km	229.82	135.70
Max Q-Factor @ 50 km	278.18	208.62
Min BER (all distances)	0	0
Eye Height @ 50 km	0.000493	0.000442
Max Reach (zero BER)	>50 km	>50 km

The comparative results highlight several scalability properties of the hybrid WDM/TDM PON design.

Linear Aggregate Capacity Growth. The aggregate capacity scales linearly with the number of wavelength channels because each channel carries an independent 1 Gbps stream and the channels share the fiber spectrum without time-domain contention. The 4 Gbps and 8 Gbps figures shown in Fig. 8 confirm this linear relation, and the same trend extrapolates naturally to 16 Gbps for a future 16-channel design on the same DWDM grid.

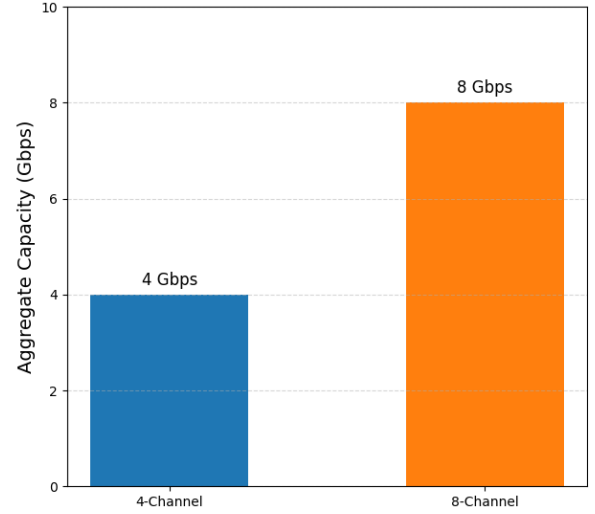


Fig. 8. Aggregate Capacity Comparison between the 4-channel and 8-channel hybrid WDM/TDM PON architectures.

Linear User Scalability. Under the $N_{\text{users}} = 2N_{\lambda}$ relation imposed by the 1×2 power-splitter design, doubling channel count exactly doubles user count, as illustrated in Fig. 9. Higher splitting ratios such as 1×4 or 1×8 would multiply user density further, at the cost of additional 6 or 9 dB of splitting loss respectively, which the present power budget margins suggest can be absorbed at moderate distances.

Bounded Q-Factor Degradation. The per-branch Q-factor of the 8-channel system is consistently lower than that of the 4-channel system, but the reduction is bounded and well-behaved. Averaging across all receiver branches, the 4-channel Q-factor falls from approximately 309 at 10 km to 251 at 50 km, while the 8-channel Q-factor falls from approximately 187 at 10 km to 170 at 50 km. Both systems remain more than an order of magnitude above the $Q \geq 6$ threshold even at the worst-case branch. The channel-pair structure of the OptiSystem WDM filters means that the outermost wavelengths benefit from less inter-channel spectral overlap and consistently record the highest Q-factors, while the mid-band wavelengths record the lowest, but every branch crosses the acceptance threshold by a wide margin.

Error-Free Operation Preserved. Despite the additional spectral congestion of the 8-channel design, BER remains at zero on every receiver branch at every measured fiber length.

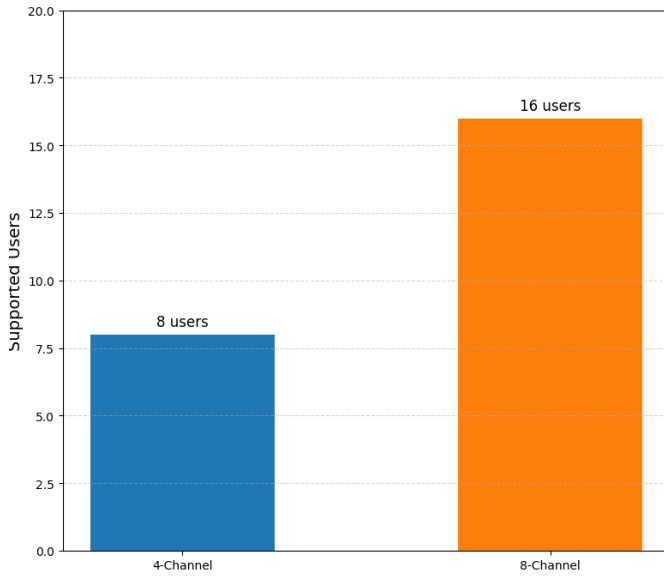


Fig. 9. User Scalability Comparison between the 4-channel and 8-channel hybrid WDM/TDM PON architectures under a 1×2 power-split per channel.

This is the most significant scalability outcome: doubling the channel count does not compromise the error-free property of the link within the tested parameter ranges. Wavelength utilization efficiency therefore improves from 4 to 8 simultaneous channels within the same C-band spectral region without any reliability penalty at 1 Gbps per channel.

Quality of Service Implications. The large Q-factor margins available in both systems leave substantial headroom for quality-of-service differentiation through dynamic wavelength and bandwidth allocation algorithms [11]. With eight wavelengths to allocate, the 8-channel system supports finer granularity in service-class scheduling than the 4-channel system, which translates directly into more flexible QoS provisioning for heterogeneous user traffic.

VIII. CONCLUSION AND FUTURE WORK

This paper presented a scalability-focused OptiSystem 7.0 simulation study of hybrid WDM/TDM Passive Optical Networks, comparing a 4-channel architecture on the 193.1–193.4 THz grid against an 8-channel architecture on the 193.1–193.8 THz grid, both operating at 1 Gbps per channel over fiber lengths of 10, 30, and 50 km. The 4-channel system supports eight end users at 4 Gbps aggregate capacity, and the 8-channel system supports sixteen users at 8 Gbps aggregate capacity.

Both architectures maintain zero Bit Error Rate on every receiver branch at every measured fiber length. Q-factors range from 247.63–385.82 (4-channel, 10 km) and 229.82–278.18 (4-channel, 50 km), and from 135.70–245.37 (8-channel, 10 km) and 135.70–208.62 (8-channel, 50 km). The 8-channel system therefore delivers double the capacity and double the user density at the cost of a moderate Q-factor reduction that remains far above the industry $Q \geq 6$ acceptance threshold.

The principal scalability findings are:

- Aggregate capacity scales linearly with the number of wavelength channels.
- Supported user count scales linearly under fixed splitting ratio.
- Q-factor reduction with increased channel count is bounded and predictable, governed primarily by multiplexer insertion loss and inter-channel passband interaction.
- BER remains at zero through the 4 $\rightarrow$ 8 channel transition across 10, 30, and 50 km.
- Wavelength utilization efficiency doubles within the same C-band region without any reliability penalty.

Future research directions include:

- Extension to 16-channel and 32-channel architectures on the ITU-T DWDM grid to test linear scaling assumptions at higher density.
- Operation at higher per-channel bit rates (10 Gbps and 40 Gbps) where dispersion and nonlinearity become non-negligible.
- EDFA-amplified long-reach extension beyond 100 km.
- Integration of dynamic wavelength and bandwidth allocation algorithms for QoS differentiation across the eight available wavelengths.
- Higher-ratio power splitters (1×4 , 1×8) to further increase user density per fiber.

ACKNOWLEDGMENT

The authors gratefully acknowledge the Department of Computer Science and Engineering, University of Asia Pacific, for providing access to the OptiSystem simulation software and laboratory facilities. The authors also thank Md. Al-Amin Sany for guidance throughout the project.

REFERENCES

- [1] G. P. Agrawal, *Fiber-Optic Communication Systems*, 4th ed. New York, USA: John Wiley & Sons, 2010.
- [2] N. Ö. Ünverdi and N. A. Ünverdi, "Modeling of Optical Communication Systems," in *Proc. V. URSI-Turkey Scientific Congress*, Guzelyurt, Northern Cyprus, Aug. 2010, pp. 436–439.
- [3] Ö. Güre, "Passive Optical Networks and Applications," M.Sc. Thesis, Yildiz Technical University, Istanbul, Turkey, 2012.
- [4] T. Koonen, "Fiber to the Home/Fiber to the Premises: What, Where, and When?" *Proceedings of the IEEE*, vol. 94, no. 5, pp. 911–934, May 2006.
- [5] İ. Uzar and N. Ö. Ünverdi, "FTTX Technology and Applications Used in Optical Communication Systems," in *Proc. ELECO 2014*, Bursa, Turkey, Nov. 2014, pp. 597–602.
- [6] G. Keiser, *FTTX Concepts and Applications*. New Jersey, USA: John Wiley & Sons, 2006.
- [7] P. E. Green, *Fiber to the Home: The New Empowerment*. New Jersey, USA: John Wiley & Sons, 2006.
- [8] S. Rajalakshmi, A. Srivastava, and A. Pandey, "Analysis of TDM and WDM PON using Different Coding Schemes for Extended Reach," *IJC-SNS International Journal of Computer Science and Network Security*, vol. 11, no. 7, pp. 120–126, Jul. 2011.
- [9] S. Jain and A. B. Therese, "Design and Performance Evaluation of Hybrid WDM/TDM Passive Optical Networks Using Star Topology," *International Journal of Computer Science*, 2014.
- [10] E. Koman and N. Ö. Ünverdi, "Analysis and Applications of the Hybrid WDM/TDM Passive Optical Networks," *Sigma Journal of Engineering and Natural Sciences*, vol. 37, no. 4, pp. 1059–1073, 2019.

- [11] A. R. Dhaini, C. M. Assi, M. Maier, and A. Shami, "Dynamic Wavelength and Bandwidth Allocation in Hybrid TDM/WDM EPON Networks," *Journal of Lightwave Technology*, vol. 25, no. 1, pp. 277–286, Jan. 2007.
- [12] ITU-T, "Spectral Grids for WDM Applications: DWDM Frequency Grid," ITU-T Recommendation G.694.1, Geneva, Switzerland, 2020.
- [13] ITU-T, "Characteristics of a Single-Mode Optical Fibre and Cable," ITU-T Recommendation G.652, Geneva, Switzerland, 2016.
- [14] Optiwave Systems Inc., *OptiSystem: Optical Communication System Design Software*, User Documentation. Ottawa, Canada: Optiwave, 2020.